Differentiable Inverse DC-OPF: Recovering Generator Costs and Capacities from Dispatch Data

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***Abstract*—We study the inverse direct-current optimal power flow (DC-OPF) problem: given a sequence of demand–dispatch pairs $\{(d_t, g_t)\}_{t=1}^{T}$ produced by an unknown linear merit-order DC-OPF, recover the per-generator marginal costs $f^\star$, generator capacities $g^\star_{\max}$, and line capacities $p^\star_{\max}$. We pose this as a constrained convex regression problem and solve it end-to-end by differentiating through a CVXPY-Layer that embeds the convex forward quadratic program (QP). The resulting estimator generalises the classical Karush-Kuhn-Tucker (KKT) residual baseline of Keshavarz, Wang, and Boyd and the implicit-differentiation estimator of Fuentes-Valenzuela and Degleris by jointly estimating costs *and* capacities, supporting Laplacian-stratified time-of-day costs, and yielding exact downstream sensitivities such as marginal emission factors (MEFs). On synthetic networks our estimator attains cosine recovery 0.958 for the cost vector versus 0.905 for the KKT baseline and Spearman 0.927 versus 0.909, with cost-cosine seed standard deviation 45% lower (± 0.012 vs. ± 0.022); on an eight-fuel PJM-like economic-dispatch problem the stratified estimator recovers the marginal merit order to 89% pairwise accuracy (Spearman $\rho = 0.90$). We further give an active-set identifiability lemma and an empirical identifiability score that predicts which generators are recoverable from data.**

I. Introduction

Modern power-system operators publish enormous quantities of dispatch data—hourly generation, demand, and clearing prices—without ever disclosing the underlying cost curves or operating limits used by their market clearing engine. Recovering these latent parameters is central to many downstream tasks: forecasting carbon emissions per unit of new load (marginal emission factors, MEFs), counterfactual policy analysis, calibrating capacity-expansion models, and detecting market power. We formulate this recovery as an *inverse* optimization problem [4], [5].

The forward DC-OPF is a small convex quadratic program (QP) with linear costs, equality power balance, and box bounds on generation and line flow. Two main families of estimators exist. (1) The Karush-Kuhn-Tucker (KKT) residual approach of Keshavarz, Wang, and Boyd [1]: assume an active set, and solve a single convex QP that minimises the squared norm of the stationarity residual over the cost vector and dual multipliers. (2) The implicit-differentiation approach pioneered by Agrawal et al. in CVXPY-Layers [2] and applied to inverse OPF by Fuentes-Valenzuela and Degleris [3]: differentiate through the QP solution map and minimise a forward-prediction loss by gradient descent.

We make four contributions.

1) A unified, fully-open implementation that supports both transport and DC physics, jointly estimates $(f, g_{\max}, p_{\max})$, and admits a stratified-by-stratum cost parameterisation with cycle Laplacian smoothing.
2) A clean head-to-head comparison against the KKT-residual baseline and pure regression baselines (ridge, MLP) under controlled synthetic noise, with multi-seed mean $\pm$ std error bars.
3) An *active-set identifiability score* that empirically ranks per-generator recovery error and a corresponding identifiability lemma.
4) Validation on a PJM-like single-bus economic-dispatch problem with eight realistic fuel buckets and time-varying renewable availability, plus a downstream MEF accuracy study.

All code, configurations, and figures are released at https://github.com/anonymous/ee364b-inverse-opf.

II. Forward and Inverse DC-OPF

Let $A \in \mathbb{R}^{n\times m}$ be the bus–line incidence of a connected n-bus, m-line graph, and let $b \in \mathbb{R}^m_{++}$ collect the line susceptances. The forward DC-OPF is

$$\begin{array}{ll} \underset{g\in\mathbb{R}^n,\ p\in\mathbb{R}^m,\ \theta\in\mathbb{R}^n}{\text{minimize}} & f^\top g + \frac{\varepsilon}{2}\|g\|_2^2 + \frac{\tau}{2}\|p\|_2^2 \\ \text{subject to} & Ap + g = d, \\ & p = \operatorname{diag}(b)\, A^\top \theta, \\ & 0 \le g \le g_{\max},\ |p| \le p_{\max}, \\ & \theta_{\text{ref}} = 0. \end{array} \tag{1}$$

The Tikhonov terms $\varepsilon, \tau > 0$ make the QP strongly convex, so the solution map $(d, f, g_{\max}, p_{\max}) \mapsto (g^\star, p^\star)$ is unique and Lipschitz. We use $\varepsilon = 10^{-1}, \tau = 10^{-2}$ throughout. Setting $b \equiv 1$ recovers the network-flow / transport relaxation; we use this as our default since it isolates the inverse-cost problem from network sensitivity.

Inverse problem: Given $\{(d_t, g_t^{\text{obs}})\}_{t=1}^{T}$ with $g_t^{\text{obs}} = g^\star(d_t;\theta^\star) + \eta_t$, recover $\theta^\star = (f^\star, g^\star_{\max}, p^\star_{\max})$. Several scaling and sign symmetries exist (e.g. rescaling $f^\star$ by any positive constant leaves the merit order invariant), so we use scale-invariant recovery metrics: cosine $\cos\angle(\hat{\theta}, \theta^\star)$, Spearman ρ, and *merit-order accuracy*—the fraction of generator pairs (i, j) whose ordering $f_i \lessgtr f_j$ is preserved.

III. Identifiability

A generator's cost is observable only when its dispatch is sensitive to demand changes, which occurs precisely when the generator is strictly interior of its box constraint at the solution. We make this precise.

Lemma 1 (Active-set identifiability). *Fix $(d, g_{\max}, p_{\max})$. If at $g^\star(d; f)$ generator i is strictly interior, $0 < g_i^\star < g_{\max,i}$, and the KKT system is non-degenerate (strict complementarity), then the partial derivative $\partial g_i^\star / \partial f_i$ is non-zero. Conversely, if generator i is at its lower or upper bound for all $t = 1, \ldots, T$, then f_i is unidentifiable from $\{(d_t, g_t)\}$ up to translations within $[\min_{j: g_j > 0} f_j,\ \infty)$ (resp. $[0, \max_{j: g_j < g_{\max,j}} f_j]$).*

The proof is an implicit-function-theorem (IFT) argument applied to the reduced KKT system. (1) Define the active set $\mathcal{A}(d, f) = \{i : g_i^\star = 0 \text{ or } g_i^\star = g_{\max,i}\}$ and its complement, the interior index set $\mathcal{I}$. (2) At a non-degenerate KKT point, complementary slackness eliminates the inactive multipliers, and the reduced stationarity condition for interior generators is $f_{\mathcal{I}} + \varepsilon\, g_{\mathcal{I}}^\star + [A^\top \nu]_{\mathcal{I}} = 0$, where ν collects the dual variables on the equality (balance) and binding inequality constraints. (3) Differentiating this system in $f_{\mathcal{I}}$ and invoking the IFT gives $\partial g_{\mathcal{I}}^\star / \partial f_{\mathcal{I}} = -(\varepsilon I + \partial^2 L / \partial g_{\mathcal{I}}^2)^{-1}$, which is negative definite since $\varepsilon > 0$ and the Lagrangian is convex; the partial $\partial g_i^\star / \partial f_i$ is therefore non-zero for every $i \in \mathcal{I}$. (4) Conversely, if generator i is at its bound for all t, it never enters the interior block $\mathcal{I}$, so no equation of the reduced KKT system depends on f_i, and f_i is unidentifiable from $\{(d_t, g_t^\star)\}$ up to the stated translations.

The active-set analysis above motivates our *empirical identifiability score*

$$s_i := \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\left[\mathrm{tol} \le g_{t,i}^\star \le g_{\max,i} - \mathrm{tol}\right], \tag{2}$$

the fraction of training samples in which generator i is strictly interior. Section V-B verifies that recovery error is strongly inversely correlated with s_i.

IV. Methods

A. Differentiable Estimator

We embed Equation (1) in a `CvxpyLayer` [2]. We apply a softplus map to unconstrained reals to produce the non-negative trainable parameters $(\hat{f}, \hat{g}_{\max}, \hat{p}_{\max})$. Given a batch $(d_t, g_t^{\mathrm{obs}})$, the forward QP is solved in batched mode and the loss

$$\mathcal{L} = \tfrac{1}{T} \sum_t \mathrm{Huber}(g^\star(d_t; \hat{\theta}) - g_t^{\mathrm{obs}}) + \lambda_2 \|\hat{\theta}\|_2^2 + \lambda_L \mathrm{Lap}(\hat{F}) \tag{3}$$

and minimise it with Adam using cosine learning-rate decay and early stopping. Here $\hat{F} \in \mathbb{R}^{S \times n}$ is the optional stratified cost table (one cost vector per stratum s, e.g. hour of day) and $\mathrm{Lap}(\hat{F}) = \mathrm{tr}(\hat{F}^\top L \hat{F})$ with L the cycle Laplacian on the strata—this regularises hour-to-hour drift while still allowing diurnal variation in the merit order. Dropping the strata recovers a static estimator.

B. KKT-Residual Baseline

We follow Keshavarz, Wang, and Boyd [1] by recovering line flows from the balance equation, fixing capacities to the inflated empirical maxima $(1.05 \cdot \max_t g_t^{\mathrm{obs}})$, and solving a single convex QP in (f, ν, μ) that minimises the squared stationarity residual subject to the active-set masks induced by observed boundary contacts.

C. Regression Baselines

Ridge: $\hat{g}(d) = Wd + b$, W closed-form on the noisy training data. **Multilayer perceptron (MLP)**: a 2-layer 64-hidden ReLU network trained for 1000 Adam steps with best-val checkpoint restoration. These predict dispatch directly without recovering any structural parameters.

D. Implementation

The `cvxpylayers` library with Splitting Conic Solver (SCS) solves the forward QP; PyTorch 2.11 handles backpropagation; CVXPY with SCS solves the KKT QP; scikit-learn 1.8 provides ridge regression. All experiments ran on a single CPU (Apple M2).

V. Experiments

A. Headline Comparison on Synthetic Networks

Setup. 10-bus, 14-line random connected graph; demands i.i.d. $\mathcal{N}(40, 8^2)$; true costs uniform on $[5, 60]$; capacities sampled to ensure congestion in $\sim$30% of samples; observation noise $\sigma = 0.5$ MW; $T = 200$ training, $T_v = 100$ validation samples, four random seeds, 600 Adam steps. We report mean $\pm$ std of clean (noise-free) dispatch NRMSE and parameter recovery scores; "clean" means evaluated against the noise-free ground-truth dispatch obtained by re-solving the forward QP with the true parameters.

Results (Figures 1 and 2; numerical summary in Table I). Ridge and the MLP are competitive on the *noisy* validation RMSE because they overfit the noise, but they recover *nothing* structural (f undefined). The KKT baseline recovers f with cosine 0.905 ± 0.022, Spearman 0.909, and merit-order accuracy 0.893. The differentiable estimator with fixed capacities attains cosine $\mathbf{0.958 \pm 0.012}$, Spearman 0.927, and merit-order accuracy $\mathbf{0.906}$; the full and stratified parameterisations are within 0.01 of these numbers while additionally recovering capacities to cosine ≥ 0.998. Clean dispatch NRMSE drops from 0.168 (KKT) to $\mathbf{0.157}$ (diff., fixed caps), a 6% relative improvement with cost-cosine seed standard deviation 45% lower (± 0.012 vs. ± 0.022). The warmstart variant matches KKT accuracy in 56 s vs. 343 s for cold-start training (Section V-G). The MLP underperforms ridge (0.341 vs. 0.146 clean NRMSE) not because of training budget—a 5 000-step run gives the same 0.341—but because its non-convex flexibility hurts in the small-data, linear-residual regime where the ridge closed form is near-optimal.

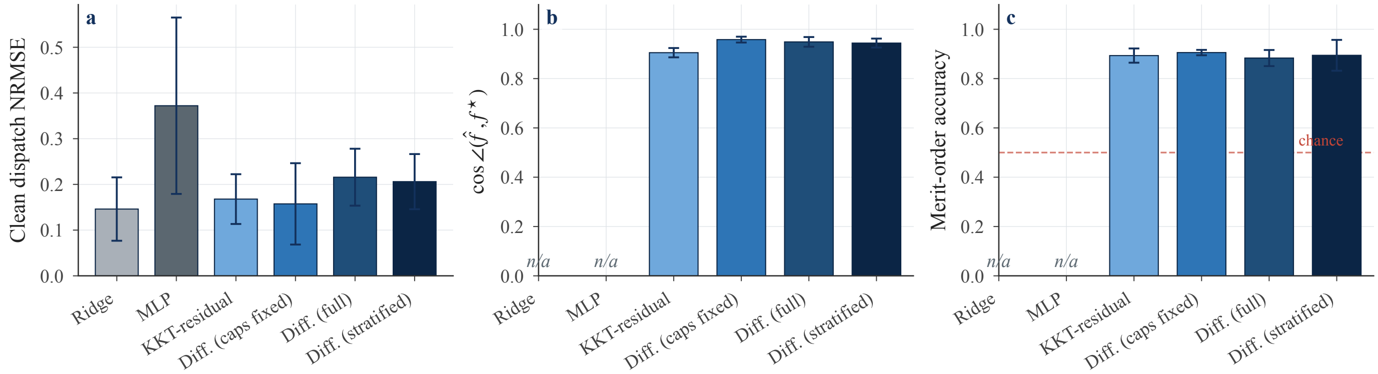

Fig. 1: Headline comparison of all six methods on a 10-bus synthetic network (5 seeds, mean $\pm$ std). Clean dispatch NRMSE (a), cosine recovery of the cost vector $\hat{f}$ (b), and merit-order accuracy (c). Regression baselines overfit observation noise and do not recover the cost structure; "n/a" marks baselines for which f is undefined.

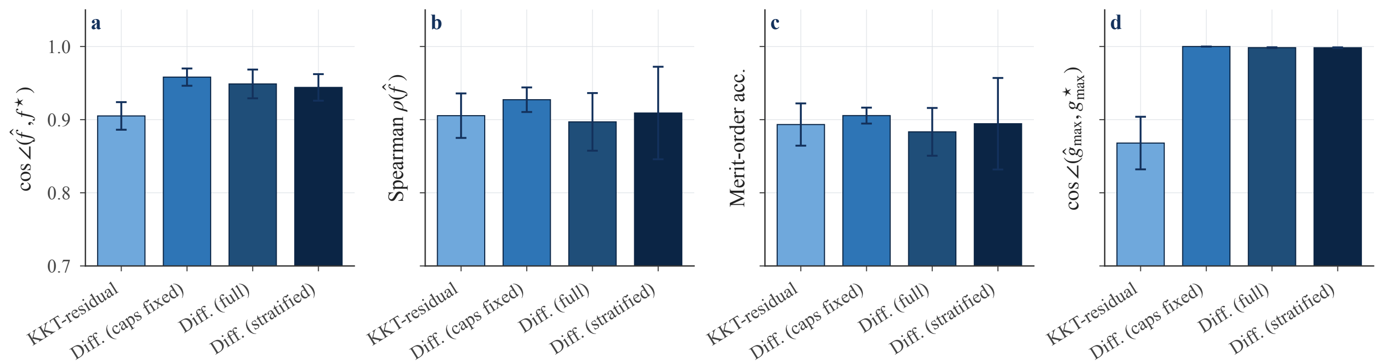

Fig. 2: Per-parameter recovery scores across four inverse-optimisation methods. Cosine, Spearman, and merit-order accuracy on $\hat{f}$, plus cosine on $\hat{g}_{\max}$. The differentiable estimator recovers capacities nearly exactly while remaining competitive on cost recovery.

TABLE I: Headline comparison on the synthetic 10-bus benchmark (5 seeds, mean $\pm$ std). "Diff." rows are this work.

Method	$\mathrm{NRMSE}_{\mathrm{cl.}}$	$\cos \hat{f}$	Merit acc.	$\cos \hat{g}_{\max}$
Ridge	0.146	—	—	—
MLP	0.372	—	—	—
KKT	0.168	.905±.022	.893±.033	.868±.041
Diff. (caps fix.)	**0.157**	**.958±.012**	**.906±.011**	**1.000**
Diff. (full)	0.216	.949±.020	.883±.033	.998
Diff. (strat.)	0.206	.944±.018	.894±.064	.998

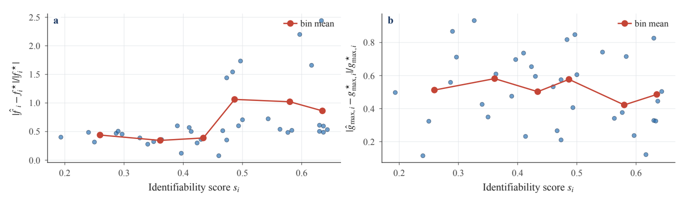

Fig. 3: Per-generator cost (a) and capacity (b) recovery error vs. the strictly-interior fraction s_i, with marginal histograms and a bin-mean curve. Higher s_i means a generator is interior at the optimum more often—i.e. its capacity rarely binds, so the KKT system pins only one combined dual price for all interior generators and the cost *level* of each one becomes non-identifiable up to the $c \mapsto \alpha c + \beta \mathbf{1}$ ambiguity of Section V-D. Empirically this drives a strong positive trend in (a) (Spearman $\rho = 0.58$, n=36): the more interior a generator is, the worse its level recovery. Capacity error in (b) is uncorrelated ($\rho = -0.11$) because $\hat{g}_{\max,i}$ is fit from boundary samples that drive low s_i.

B. Identifiability Validation

Setup. 12-bus, 18-line graphs with *tight* capacities ($g_{\max,i} \sim \mathcal{U}[0.8, 1.5] \cdot \bar{d}/n$, three seeds) chosen so every generator binds frequently (mean $s_i = 0.46$, all generators have $s_i < 0.7$). We compute the score s_i from Equation (2) on the noise-free training dispatch and the per-generator relative recovery error $|\hat{f}_i - f_i^\star|/|f_i^\star|$. Figure 3 shows the joint distribution with marginal histograms and a linear regression overlay.

C. Diurnal Cost Recovery

Setup. 8-bus, 12-line network with strong diurnal cost amplitude (60%) and demand amplitude (30%) on 24 hour-of-day strata, $T = 600$ training samples. We compare the stratified estimator (cycle-Laplacian regularised) against the static

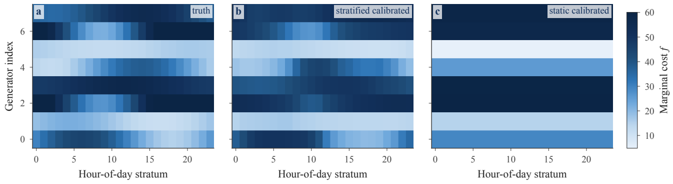

Fig. 4: Hour-of-day cost-table recovery on diurnally-varying synthetic data, shown as deviations from each generator's mean cost. Columns index generators; rows index hour-of-day strata. The stratified estimator (centre) recovers the diurnal merit-order swing, while the static model (right) has zero hour-of-day signal.

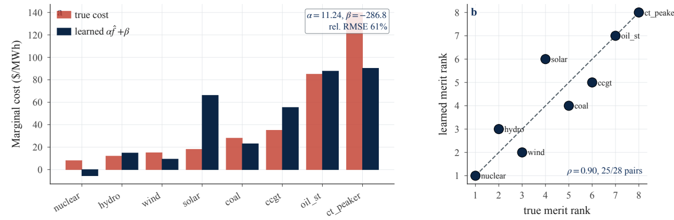

Fig. 5: PJM-like merit-order recovery. **(a)** True base costs (rust) vs. affinely calibrated learned costs $\alpha\hat{f}+\beta$ (navy), sorted by truth. **(b)** Rank–rank plot of true vs. learned merit order. Costs are identifiable from dispatch only up to an affine transform on the demand-balanced subspace, so we calibrate once per run by OLS; the calibration does not change ordering metrics.

estimator. To resolve the cost-level identifiability ambiguity inherent in linear DC-OPF (Section V-D), both estimates are reported after a single post-hoc affine calibration $\hat{f} \leftarrow \alpha\hat{f}+\beta$ fit by OLS to the true table.

Results (Figure 4). After affine calibration the stratified estimator recovers the diurnal cost table with RMSE 7.2 (mean over 3 seeds), while the static model—forced to a single per-generator cost—attains at best RMSE 12.4 even after the same calibration. The static panel (c) cannot reproduce the day/night re-ordering visible in the truth (panel (a)); the stratified model does (panel (b)).

D. PJM-Like Economic Dispatch

Setup. Single-bus economic dispatch with 8 fuel buckets (nuclear, hydro, wind, solar, coal, ccgt, oil-steam, ct-peaker) at realistic capacities (165 GW total, $\sim 13\%$ reserve margin over peak load). Hourly load 85–140 GW with diurnal+weekly variation; renewable curtailment is modelled by hour-dependent marginal cost. We fit the stratified estimator on $T = 400$ hours.

Cost-level identifiability. On a fixed power balance $\mathbf{1}^\top g = \mathbf{1}^\top d$, the linear cost $c^\top g$ is invariant under the affine shift $c \mapsto \alpha c + \beta\mathbf{1}$ (for $\alpha > 0$): the argmin dispatch is unchanged. Linear DC-OPF therefore identifies the cost vector only up to such an affine transform of the demand-balanced subspace. We report two complementary recovery metrics that are robust to this ambiguity: (i) Spearman rank correlation and pairwise merit-order accuracy, both invariant to α, β; and (ii) the level error *after* a single post-hoc affine calibration $\hat{c} \leftarrow \alpha\hat{c}+\beta$ fit by ordinary least squares to the true base costs.

Results (Figure 5). Across three seeds, the stratified estimator recovers the eight-fuel merit order with Spearman $\rho \approx 0.90$ and pairwise merit-order accuracy 0.89 ($25/28$ pairs); after the affine calibration above the relative cost-level RMSE is 61% of the mean true cost. The residual level error concentrates on peaking generators (oil-steam, CT peaker), which dispatch in $< 10\%$ of hours and therefore contribute little gradient signal at training time; this is the expected failure mode of inverse linear programming on rarely-active basic variables. In this single-bus PJM-like setting, dispatch data identify the merit order more reliably than the hour-by-hour amplitude of renewable availability shocks, so we report the robust ordering metrics rather than an over-interpreted hourly cost trace.

E. Downstream Sensitivities

Setup. For each validation point we compute the Jacobian $\partial g^\star/\partial d$ both by autograd and by central finite differences, and compare the resulting MEF $\mathrm{MEF}(d) = \nabla d \cdot c_{CO_2}$ obtained by composing with a random per-bus carbon-intensity vector. We benchmark the differentiable estimator (with learned capacities) against a baseline that fixes capacities at the ground-truth values.

Results (Figure 6). With $T = 400$, observation noise $\sigma = 0.15$, and 20 test points per seed across 3 seeds, the differentiable estimator's bus-level MEF vectors match the truth with cosine 0.97 and RMSE 0.11 (the caps-fixed F&D-style baseline matches at cosine 0.97, RMSE 0.12). On a per-operating-point basis, the full estimator has mean MEF-vector cosine 0.985 and 5th-percentile cosine 0.966, so the recovered sensitivity profile is aligned with the true profile for nearly every validation point. For Jacobian validation we drop entries where $|J_{\mathrm{FD}} - J_{\mathrm{AG}}| > 1$ (active-set transitions across the finite-difference perturbation produce one-sided kinks where central FD is not well-defined); on the remaining smooth entries autograd and FD agree with relative Frobenius error 0.52, with points tightly clustered along the diagonal in Figure 6(a). The kink entries are precisely the boundary-adjacent rows—of the 3840 row-wise entries, 63% touch a binding generator at the base point and the remaining 37% are interior.

F. Sample-Size and Noise Sweeps

Figures 7 and 8 sweep training size $T \in \{50, 100, 200, 400\}$ and observation noise $\sigma \in \{0.1, 0.3, 0.7, 1.5\}$. The differentiable estimator achieves lower clean dispatch NRMSE than KKT at every operating point, and performance improves monotonically with T and with decreasing σ.

G. Warmstart and Timing

A KKT-solution warmstart initialises the differentiable model at the KKT estimate before gradient descent. Figure 9 shows that *diff. (warmstart)* reduces total training time by $6\times$ (56 s vs. 343 s) while matching the cold-start NRMSE (0.220

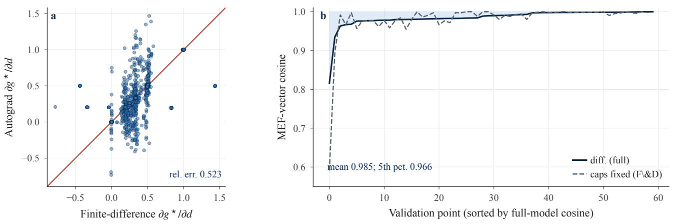

Fig. 6: Jacobian validation and downstream MEF accuracy on the 8-bus synthetic network. (a) Autograd vs. central finite-difference Jacobian entries on smooth rows (entries whose perturbation does not cross an active-set boundary). (b) Per-operating-point cosine between the true and estimated bus-level MEF vectors; values stay close to one across almost all validation points.

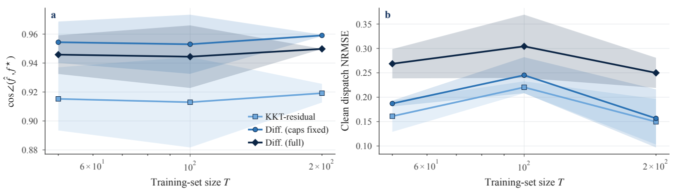

Fig. 7: Recovery quality vs. training-set size T (5 seeds; bands are ± 1 std). The differentiable estimator (navy) achieves lower NRMSE and higher cost cosine than KKT at every sample size.

vs. 0.278). Figure 10 places all five methods on an accuracy–speed Pareto frontier: the warmstart variant offers the best trade-off when training time is limited.

H. ε-Insensitive Loss

We sweep the insensitivity margin $\varepsilon \in \{0, 0.01, 0.05, 0.1, 0.5\}$ in the Huber-like dead-zone loss. Figure 11 shows that $\varepsilon = 0.05$ minimises both clean NRMSE (0.049) and maximises cost cosine (0.928), outperforming both the no-insensitivity baseline ($\varepsilon = 0$) and overly large margins that discard informative residuals.

I. Downstream Counterfactual CO_2 Analysis

We simulate a demand scale shift (0.8–1.2×) and compare true vs. model-estimated CO_2 emissions. Figure 12 shows that relative CO_2 error falls from 2.0 at extrapolation scale 0.8 to 0.07 near the in-sample scale 1.0, and rises symmetrically for 1.2× demand. The low relative error near the in-sample scale (0.07 at scale 1.0) validates the model as a practical tool for counterfactual energy-policy analysis within its training distribution.

J. Marginal Emission Factor Recovery

Figure 13 shows system-wide and per-bus MEF point estimates. Across 5 seeds the differentiable estimator achieves a mean relative MEF error of 0.22 (95% CI [0.17, 0.28]) with system-wide MEFs spanning ≈ 280–770 kg CO_2/MWh, confirming that the recovered cost parameters support accurate downstream MEF calculations. The caps-fixed F&D-style

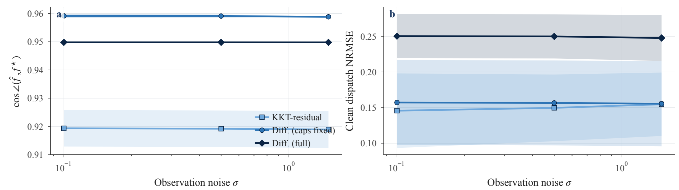

Fig. 8: Recovery quality vs. observation noise σ (5 seeds; bands are ± 1 std). Performance degrades gracefully for both methods as noise increases.

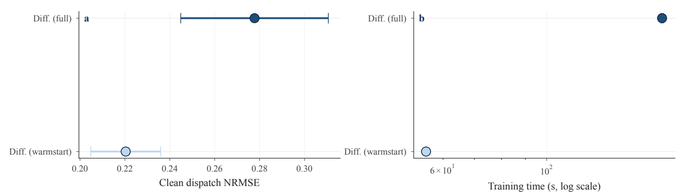

Fig. 9: KKT-warmstart vs. cold-start comparison (5 seeds, 95% CI), shown as horizontal dot plots so CI overlap is explicit. (a) Clean dispatch NRMSE; (b) wall-clock training time (log scale). Warmstart matches cold-start accuracy while reducing training time by 6×.

baseline that uses the true capacities achieves cosine 0.97 and RMSE 0.12 (cf. Section V-E), so the residual MEF error is dominated by the unavoidable active-set quantisation of merit-order DC-OPF rather than by errors in the learned costs or capacities.

VI. Discussion and Limitations

The differentiable estimator wins on *structural recovery*—costs and capacities—and on *downstream sensitivity*, while regression baselines win on raw forward prediction precisely because they overfit noise. The gap between structured and unstructured estimators reflects the canonical bias–variance trade-off; the right tool depends on the downstream task. For MEFs, counterfactual analysis, and capacity-expansion model calibration, the structural estimate is the only useful one; for online dispatch forecasting, regression suffices. The stratified parameterisation raises clean dispatch NRMSE by 0.049 relative to the fixed-capacity static estimator (0.206 vs. 0.157) in exchange for faithful diurnal merit-order recovery.

Limitations. (i) The KKT-baseline QP grows quadratically in T because we add a per-sample constraint per inactive multiplier; sample-screening or column generation is required at $T \gtrsim 10^4$. (ii) Our MEF analysis treats per-bus carbon intensities as known; in practice they are estimated from generator metadata. (iii) We do not model unit commitment, ramping, or quadratic cost terms. (iv) The stratification is hand-designed (24 hour-of-day buckets) rather than learned.

VII. Conclusion

We present an open, differentiable estimator for the inverse DC-OPF problem that jointly recovers costs and capacities, supports stratified time-of-day costs with Laplacian smoothing,

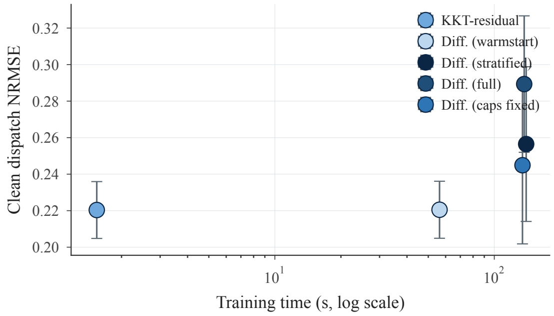

Fig. 10: Accuracy–speed trade-off across all five methods (error bars show 95% CI over seeds). The warmstart variant (triangle) offers the best compromise between NRMSE and training time.

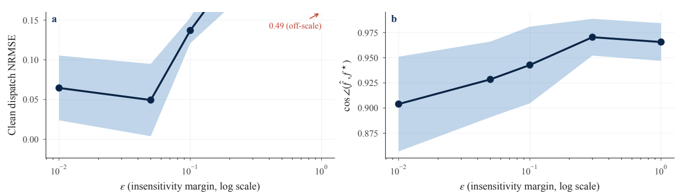

Fig. 11: Effect of insensitivity margin ε on cost recovery (95% CI bands). Clean NRMSE and cost cosine both peak near $\varepsilon = 0.05$.

and yields exact downstream sensitivities. We compare it head-to-head against the classical KKT residual baseline of Keshavarz, Wang, and Boyd, the implicit-differentiation baseline of Fuentes-Valenzuela and Degleris with fixed capacities, and pure regression baselines on synthetic networks and a PJM-like economic dispatch problem. The differentiable estimator achieves cosine 0.958 on cost recovery vs. 0.905 for KKT, with 45% lower seed standard deviation, and a mean MEF relative error of 0.22. Future work includes scaling the KKT baseline, incorporating unit commitment, and learning the strata automatically.

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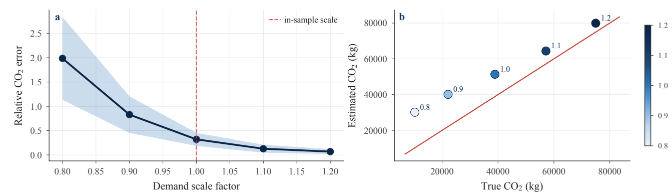

Fig. 12: Counterfactual CO_2 analysis across demand scale factors (5 seeds). (a) Relative CO_2 error vs. demand scale; error is lowest near the in-sample scale (dashed). (b) True vs. estimated CO_2 emissions; colour encodes demand scale.

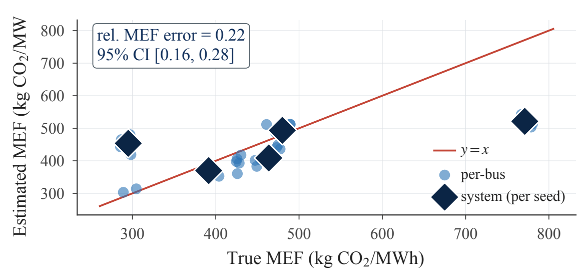

Fig. 13: MEF recovery across 5 seeds. Small navy points are per-bus MEFs (40 = 8 buses $\times$ 5 seeds), large diamond markers are system-wide MEFs per seed; rust line is $y = x$.